

AI-Driven Data Fusion for Humanitarian Visual Analytics: Integrating Multimodal Social Media Insights on Ukrainian Refugee Displacement

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Abstract—The ongoing displacement of Ukrainian refugees into Poland highlights critical gaps in timely, data-driven humanitarian response. This paper presents Aegis, a visual analytics platform that integrates large language models (LLMs) to extract, structure, and analyze social media narratives in real time. A case study-based user evaluation and testing of Aegis with humanitarian professionals assessed Aegis’s ability to identify service access barriers and community concerns. Results showed that Aegis reduced analytic burden, enabled rapid thematic discovery, and supported actionable insights. While users found the system intuitive and effective, they emphasized the need for improved geolocation accuracy, interface flexibility, and AI transparency. Recommendations include refining models, scaling data capacity, and deploying Aegis in live crisis settings. This work contributes to humanitarian technology by demonstrating how LLM-driven tools can enhance situational awareness and support evidence-based decision-making in forced displacement contexts.

Index Terms—Humanitarian technology, Visual analytics, Natural language processing, Social media analysis, Forced migration, Refugee informatics, Humanitarian decision support

I. INTRODUCTION

The displacement of Ukrainian refugees into Poland remains an ongoing humanitarian challenge nearly three years after Russia’s full-scale invasion. A persistent issue in crisis response is developing effective analytical procedures to support evidence-based decision-making in resource-constrained environments. Traditional methods of monitoring displacement often rely on fragmented datasets and formal reporting structures, which may not capture real-time, community-driven insights. Social media platforms, however, provide a rich source of heterogeneous open-source information—text, images, and video—that can offer valuable perspectives on refugee experiences, public sentiment, and emergent needs. Yet transforming this unstructured, multilingual content into

actionable intelligence remains a major technical and operational hurdle for humanitarian actors.

In response to this challenge, this paper presents research on the development and evaluation of Aegis, a visual analytics tool designed to extract, structure, and analyze Ukrainian refugee narratives in Poland from social media data. Building on earlier work that demonstrated the feasibility of automated translation, topic clustering, and prototype visualization, Aegis integrates large language models (LLMs) and geovisual interfaces to support humanitarian sensemaking at scale.

To assess the effectiveness of Aegis, a case study-based, task-oriented user study was conducted in collaboration with humanitarian professionals. This evaluation was guided by two key research questions: (1) How can a visual analytics platform support the identification of barriers faced by refugees in accessing essential services such as healthcare, education, employment, and housing? (2) How can such a platform help illuminate frequently asked questions among refugee communities regarding legal procedures, governmental support, and humanitarian aid?

These research questions reflect critical challenges faced by organizations working on the ground and align with documented concerns in recent literature. The evaluation also pursued two goals: a tool-centered goal to assess the technical performance of Aegis (including LLM-driven classification, geolocation, and geospatial visualization), and a user-centered goal to examine how humanitarian professionals—particularly those affiliated with the UN International Organization for Migration (IOM)—could apply the platform to gain real-time insights from public discourse.

This paper contributes to the humanitarian technology literature by demonstrating how integrated LLMs and visual analytics systems can enhance situational awareness, reduce analytical burden, and support data-driven decision-making in

forced displacement contexts.

II. METHODOLOGY

A. Prior Work

This section defines prior work on using LLMs in humanitarian analytics. Tomaszewski et al. (2024) provided key findings that serve as the foundation for the follow-up work presented in this paper [1]. In the 2024 study, the utility of LLMs, specifically ChatGPT, was evaluated in the context of developing situation awareness related to the forced displacement of Ukrainian refugees into Poland with working humanitarian practitioners. Text messages from the Help for Ukrainians in Poland Telegram group [2] were analyzed to assess the LLM’s capabilities in translating messages in multiple languages into English and identifying underlying topics and themes. Through the application of K-means clustering and word clouds, the study revealed trends and themes present at the outset of the Ukraine conflict in 2022. Evaluation of the language identification and translation functions yielded a high level of agreement among human evaluators, with a Kappa score of 0.86 and a BLEU score of 0.46, indicating effective performance of the LLM. Importantly, a key finding was that when guided by carefully constructed language prompts, LLMs are capable of processing large volumes of multilingual data with minimal human oversight—highlighting their potential as tools for humanitarian analytics in rapidly identifying critical needs, trends, and geographic locations relevant to displaced populations.

There is a growing body of literature exploring the integration of LLMs into visual analytics systems, particularly for enhancing data exploration and human-AI collaboration. Zhao et al. (2024) propose the LEVA system, which leverages LLMs to interpret visualization designs and generate insights based on system status and data, thereby supporting mixed-initiative data exploration [3]. Hutchinson et al. (2024) further expand on this direction by outlining the integration of natural language interaction directly into visual analytic workflows, highlighting both opportunities and implementation challenges in such systems [4]. In the humanitarian domain, the Humanitarian Data Insights Project (HDIP) emphasizes the accessibility and augmentative potential of GenAI tools for streamlining data workflows, arguing that these tools are easy to consume, customize, and enhance human capabilities [5]. Complementing this, Harris (2024) explores the use of generative AI for summarizing humanitarian “Situation Reports” from Relief Web, aiming to reduce cognitive load and enable rapid access to key operational information [6].

Building on these results and related work, the current paper extends the research in several key directions to produce more mature outcomes into practical humanitarian applications through the Aegis tool, an AI-driven visual analytics platform for social media analytics in crisis response.

This work contributes four specific advances: (1) the evaluation of LLMs in resource-constrained humanitarian contexts; (2) the development of a taxonomy-driven approach for classifying social media data related to displacement; (3) identifica-

tion and geocoding of location information embedded within messages, enabling the generation of geospatial visualizations that offer a more granular understanding of refugee movements and spatially distributed needs over time, and (4) a task-analysis user study assessing how LLMs and visual interfaces support analytical workflows for humanitarian practitioners. Together, these contributions demonstrate how AI-enabled data fusion and visualization can enhance crisis analytics, offering scalable solutions to support humanitarian organizations operating under growing financial and operational pressures.

B. Aegis Visual Analytics

Analytical tools were integrated into a visual analytics application called Aegis, which leverages Tableau Software to create its visual interfaces—an industry-standard platform widely used within the United Nations system [7]—as shown in Fig. 1.

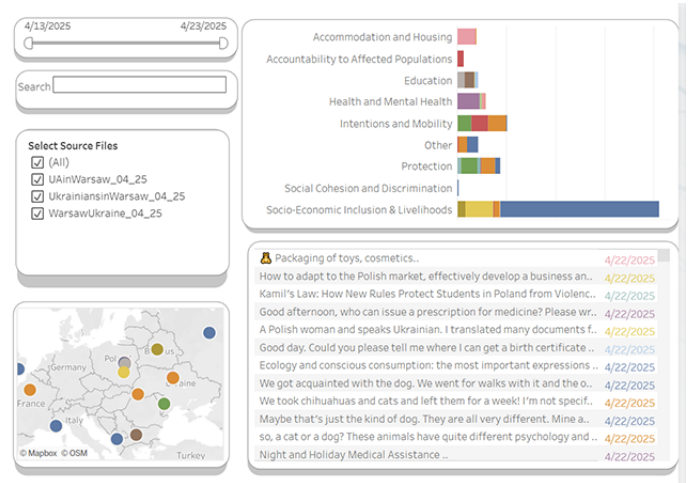


Fig. 1. Aegis Visual Analytics Dashboard

Aegis functionality includes (a) Social Media Classification and Visualization (middle of Fig. 1), (b) Social Media Geocoding (bottom left of Fig. 1), (c) AI Chat Bot Data Interaction (right of Fig. 1), and (d) AI-driven Summary Report Generation (top right of Fig. 1). The following sections describe these functionalities in greater detail.

a) Taxonomy-based Social Media Classification and Visualization: An important aspect of Aegis functionality is the ability to download a Telegram JSON data source and have it processed by artificial intelligence for translation and analysis (not shown in Fig. 1). OpenAI is utilized to translate message text into English. Image files and video frames are analyzed by OpenAI to produce a text description of the image content. Video transcriptions are gathered and translated. Once Telegram data sets have been processed they then become available for taxonomy based classification. A vector model was used to categorize messages based on the mission provided taxonomy using similarity based on [8]. The taxonomy contains parent categories, child categories, and keywords (Fig. 2).

To improve semantic classification in the Aegis platform, we drew on recent advances in natural language processing

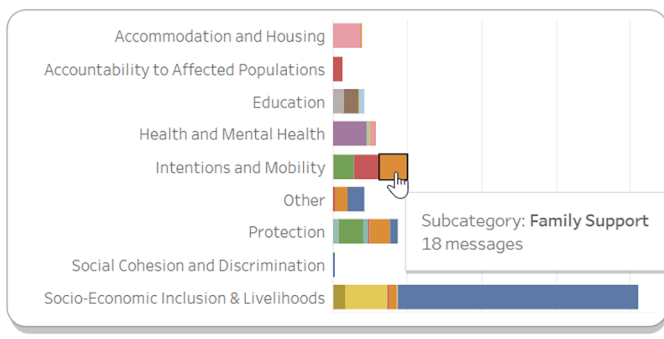


Fig. 2. Detail of the Aegis category classification and visualization

(NLP), particularly the use of Sentence Transformer models for semantic similarity. Prior work evaluating four pre-trained models on a corpus of 6,110 scientific articles—of which only 24 were relevant—found that all-mpnet-base-v2 consistently outperformed others in precision and recall, ranking relevant items within the top 700 results [9]. This evidence informed our selection of all-mpnet-base-v2 for its strong performance on noisy, real-world text.

Complementary insights were taken from Konstantinidis et al. [10], who successfully aligned user-generated YouTube tags on medical education videos to a standardized taxonomy. By tagging content with standardized terms, their tool enhanced semantic search performance and retrieval relevance. Aegis applies a similar approach to humanitarian communications by tagging message content—including translations, audio transcriptions, and photo/video summaries—with categories and subcategories aligned to a mission-specific taxonomy.

As shown in Fig. 2, messages are visualized as a horizontal bar graph, with categories labeled on the left and interactive subcategories nested within. For example, selecting “Intentions and Mobility” reveals the related subtopic “Family Support”.

To classify messages, we fine-tuned the all-mpnet-base-v2 model from the Sentence Transformers library using mission-specific parent and child categories. The model generates numerical embeddings for these categories, storing them for reference. As new messages are received, the model encodes them into embeddings and computes cosine similarity scores against the stored category embeddings. The top-matching parent and child categories are assigned to the message, along with a confidence score. Only classifications with confidence scores above 0.8 are displayed in the Aegis interface, ensuring high relevance.

This embedding-based matching approach enables efficient, scalable, and semantically rich classification of multilingual and multimedia humanitarian data aligned to evolving mission needs.

b) Social Media Geocoding: Aegis’ geocoding process is informed by recent research in NLP and geospatial inference, particularly the study by Serere, Resch, and Rudolf [11]. Their work demonstrated the effectiveness of using SpaCy and GeoPy to extract and geolocate place names from social media content. This study is especially relevant to our application, as

humanitarian message data often shares the informal and variable structure of social media posts. The authors highlighted the reliability of SpaCy’s contextual and syntactic capabilities in identifying locations, followed by GeoPy’s ability to assign latitude and longitude using the Nominatim API.

Aegis uses two Python libraries—SpaCy and GeoPy—to identify and geocode location references in messages. The process begins with SpaCy, an open-source NLP library that provides named entity recognition (NER). A fine-tuned version of SpaCy’s `en_core_web_sm` model, trained on mission-specific data, is used to extract geographical entities. To improve recall, results are also passed through the base `en_core_web_sm` model to capture any missed locations. Identified location entities are assigned confidence scores, and only those scoring above 0.8 proceed to the geocoding stage.

GeoPy, a Python client for various geocoding services, is then used to convert text-based location names into geographic coordinates. Specifically, the Nominatim API from OpenStreetMap is used to retrieve latitude and longitude for each high-confidence location tag.

Geocoded messages are visualized in a map interface, where each point’s color represents its classified category. Users can interact with the map to explore message content and geographic context, enhancing situational awareness in humanitarian response scenarios (Fig. 3).

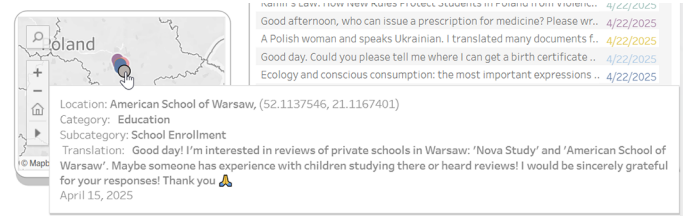


Fig. 3. Detail of the Aegis map interface

c) AI Chat Bot Data Interaction: Following industry standards of utilizing LLM interaction, an AI-driven chatbot was included with Aegis to allow end users to use open prompt engineering with ChatGPT to ask questions of the data set to supplement interaction with the previously discussed visual interfaces (Fig. 4).

The processed dataset is sent as a prompt to OpenAI along with the first user prompt, allowing for a response based directly on the data to be returned. The dataset is subsequently passed in with future prompts to take any added data into account and ensure that the dataset is within OpenAI’s context window.

d) AI-driven Summary Report Generation: Aegis also includes the ability to generate an executive summary report of data sets utilized in the visual interface. The motivation behind this was to follow humanitarian information practice of having data summaries that could be used by decision makers. In this case, LLM AI is used to create a summary of all key points of the data sets (Fig. 5).

As shown in Fig. 5, the report highlights the most frequently identified categories and locations, along with AI-generated summaries that highlight the most common questions and

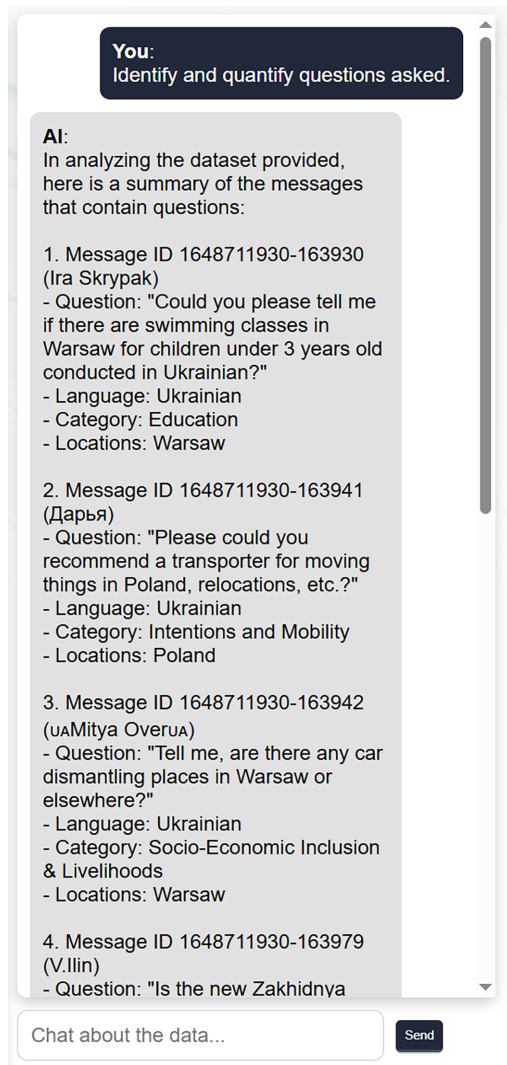


Fig. 4. Detail of the Chat Bot interface

inferred themes based on message frequency and content analysis.

C. Task-Analysis User Study

To evaluate the Aegis platform, a case study-based, task-oriented user study was developed in collaboration with working humanitarian professionals. The case study was grounded in two key analytical challenges commonly faced by humanitarian actors working with refugee populations:

- 1) Identifying barriers refugees face in accessing essential services such as healthcare, education, employment, and housing.
- 2) Understanding frequently asked questions among refugee communities related to legal procedures, governmental support, and humanitarian aid.

This case study supported two evaluation goals for Aegis:

- 1) System Performance and Technical Capability (Tool-Centered Goal): Evaluate the platform's underlying capabilities, including its AI-enabled taxonomy-driven

Data Analysis

Generate Summary Report

Get a comprehensive analysis of your data highlighting key patterns, locations, and insights.

Generate Summary

Date Range: 01/07/2025 - 03/10/2025

Top Categories:

1. Socio-Economic Inclusion & Livelihoods
2. Intentions and Mobility
3. Protection

Top Locations:

1. Warsaw
2. Poland
3. Ukrainian House
4. Gdansk

Top Questions:

1. What are the options for swimming classes for children under 3 years old conducted in Ukrainian in Warsaw?
2. Are there any car dismantling places in Warsaw or elsewhere for specific vehicle parts?
3. How to travel from Poland to Ukraine with pets?
4. Are the new train station facilities complete in Warsaw, and are trains still using the old station?

Data Inferences:

- The data reflects strong communication and support needs among Ukrainian communities, especially those residing in or transitioning to and from Poland.
- Socioeconomic inclusion is a predominant theme, emphasized by inquiries about language proficiency, financial services, and community support systems. This indicates a significant need for informational and practical support to aid in integration and livelihood establishment in host regions.
- Intentions and Mobility encompass both movement within Poland, as seen in inquiries about transport logistics, and readiness to support pets during relocation. This implies a dynamic and mobile population seeking better living conditions.
- The mention of Protection, particularly Family Support, suggests a focus on stable family environments and striving for emotional and social cohesion during relocation.
- There is a notable interest in learning resources and cultural engagements, such as folk choreography and language skills, implying a desire to maintain cultural ties while adapting to new environments.
- The data also highlights a call for financial aid or improvements in socio-economic status, reflected through discussions about benefits and banking services.
- Locations like Warsaw and Gdansk, as well as institutions like the Ukrainian House, are vital hubs for these communities, indicating where assistance and resources might be concentrated.
- Overall, the data suggests that maintaining open communication channels, providing localized support, and enhancing socioeconomic integration programs are pivotal for supporting these communities effectively.

Fig. 5. Aegis AI-driven Data Summary Report

classification of social media content, automatic location extraction and geocoding, and the use of interactive geospatial visualizations. This aimed to determine whether Aegis could improve analytical workflows beyond traditional qualitative or manual methods.

- 2) Humanitarian Insight and Decision-Making (User-Centered Goal): Assess how regional humanitarian professionals—particularly those affiliated with the International Organization for Migration (IOM)—could leverage Aegis to gain real-time insights from public discourse (e.g., Telegram messages). The goal was to determine whether Aegis could support early identification of emerging needs and inform targeted, data-driven interventions.

a) *Participant Recruitment and Study Procedures:* Usability and utility studies of the Aegis platform were conducted online during the spring of 2025. These studies had two primary objectives aligned with the aforementioned evaluation goals. The first objective was tool-centered, aiming to evaluate the technical capabilities of Aegis, including its AI-driven data fusion techniques, taxonomy-based classification, location extraction and geocoding, and geospatial visualizations supporting analytical workflows. The second objective was user-centered, focusing on how humanitarian professionals—specifically those affiliated with the International Organization for Migration (IOM)—could use Aegis to derive actionable insights from public discourse, particularly Tele-

gram messages, to identify emerging needs among displaced populations and support targeted humanitarian interventions.

The usability study focused on identifying interface-level issues that might interfere with user performance or the extraction of humanitarian insights. In contrast, the utility study assessed how effective Aegis was at supporting the reasoning processes of humanitarian professionals and its overall usefulness for operational decision-making.

All tests were conducted remotely using Zoom. Participants' sessions were screen recorded and followed by one-on-one interviews. All participants signed informed consent forms and the entire study was reviewed and approved by the Institutional Review Board (IRB) for human subjects protection at the Rochester Institute of Technology (RIT).

The usability study involved two graduate students who were recruited through email solicitation and participated voluntarily. These participants were selected to help identify any interface issues that could impede the utility study, which followed. The utility study involved three working humanitarian professionals, also recruited via email outreach. These participants evaluated how well Aegis supported humanitarian decision-making processes in real-world contexts. Although this group was primarily engaged to assess utility, they also incidentally uncovered usability concerns, which are discussed in the future work section. Due to changes in the humanitarian sector in the spring of 2025, it was difficult to obtain a large pool of working humanitarian subject participants for the study. However this situation is mitigated by the fact that each participant had an average of six years of experience in the humanitarian domain and were able to spend over 1 hour each working with Aegis to provide valuable insight, there by providing ecologically-valid insight of real end users of visual analytics tools in the humanitarian domain.

Study participants were tasked with using Aegis to analyze Telegram data to produce a short report capturing actionable insights as per the user centered goal. Specific tasks study participants were assigned using Aegis:

- 1) Interpret data processing results using visual interfaces (ranking lists widgets, date range, key word search (string match), graphic;
- 2) Use the AI open prompt to identify key barriers refugees face in accessing services, education, jobs, healthcare and other relevant topics;
- 3) Identify frequently asked questions about legal procedures, accessing governmental support, aid which can help develop targeted campaigns;
- 4) Use the map to identify the location where the main topic of conversation is Accommodations and Housing;
- 5) Identify messages within the last 3 days that mention "insurance";
- 6) Review the AI-generated summary report to summarize the data.

b) Data Collection and Qualitative Analysis Framework:

Approximately five and a half hours of screen recording data were collected from five participants. These recordings included both video and audio of participants interacting with

the system using a think-aloud protocol. In addition, follow-up interviews were conducted to elicit responses related to the two overarching research goals: (1) evaluating system performance and technical capability (Tool-Centered Goal), and (2) understanding how the tool supports humanitarian insight and decision-making (User-Centered Goal). Screen recordings were saved as MP4 files. The audio tracks were extracted and transcribed using Adobe Premiere software, and the resulting transcripts were exported as plain text files for further analysis.

To analyze the interview and think-aloud data collected for both research goals, we employed a thematic analysis approach supported by prompt-engineered interactions with ChatGPT (GPT-4). This approach was chosen to enhance the consistency, transparency, and efficiency of qualitative coding, particularly given the relatively small but information-rich sample.

For the Tool-Centered Goal, we applied a deductive coding framework grounded in established literature on visual analytics system evaluation [12] [13] [14]. This predefined framework included five core categories:

- 1) Usability and Learnability – The ease with which users could operate the system and become proficient.
- 2) Cognitive Load – The mental effort required to use the system effectively.
- 3) Trust and Credibility – Users' confidence in the accuracy and reliability of the system's outputs and data sources.
- 4) Emotional Engagement – Emotional reactions elicited by the system's content, design, or interactions.
- 5) Interpretability and Transparency – The clarity with which users understood the system's internal processes and outputs.

In contrast, for the User-Centered Goal, we conducted an inductive thematic analysis of interview transcripts to examine how participants used the system to interpret public discourse data (e.g., Telegram messages) and inform real-time humanitarian decision-making. In the initial round, open coding was applied to identify emergent concepts across the dataset. These codes were iteratively refined and grouped into broader thematic categories, culminating in the development of a structured codebook. While some themes aligned with the semi-structured interview questions—reflecting our User-Centered Goal—others emerged organically from participants' unprompted reflections and their hands-on interaction with the tool. The analysis was informed by conceptual frameworks in humanitarian informatics and crisis analytics [15] [16] [17] [18], but remained grounded in participant responses. This hybrid approach allowed us to capture both anticipated and novel insights relevant to the use of AI-driven visual analytics in humanitarian contexts.

c) Qualitative Analysis Prompt Engineering Methodology: For both goals, qualitative analysis was conducted using ChatGPT-driven prompt engineering, following the structured design principles outlined by Zhang et al. (2025) [19]. Each prompt included study context, clear task instructions for thematic coding, guidance on data segmentation and code

extraction, a defined output format (tables or lists with quotes), and optional enhancements like summarization or theme clustering.

This unified methodological pipeline allowed us to generate thematically rich, interpretable outputs while ensuring traceability and consistency in how data were analyzed across the two research goals. The use of ChatGPT in this workflow was iterative and reflexive; outputs were reviewed and refined by researchers to preserve interpretive nuance and ensure alignment with the study’s conceptual framing.

III. RESULTS AND DISCUSSION

A. Usability Testing with Non-Humanitarian Experts

To ensure that no usability issues would interfere with the two aforementioned research goals, the Aegis tool was first evaluated with two non-humanitarian experts prior to testing with humanitarian professionals. This initial usability testing allowed for iterative refinements and provided early insights into user perceptions across several key dimensions.

The aforementioned Tool-Centered Goal deductive coding framework was applied to participant feedback given the framework’s relevance to system usability. Overall, both participants found the system intuitive and accessible. As one participant noted, coded under *Usability and Learnability*:

“This is all, like, really intuitive, really usable, easy to understand.”

Similarly, another user echoed this sentiment:

“It was easy... only part where I was a little confused was the message still showing after I clicked on it.”

Despite overall ease of use, some mild *Cognitive Load* challenges were observed, particularly with unexpected visual persistence in interface elements. For instance:

“I’m like, oh, why is it still showing up?... do I click it again?”

This indicated a need for clearer affordances and feedback mechanisms in the interface.

Under *Trust and Credibility*, one participant raised questions about data handling:

“It doesn’t give out names of people... what’s the data pulling from?”

This highlights the importance of transparent data provenance, even among non-expert users.

Participants also described positive emotional responses to the experience, which was reflected in *Emotional Engagement* coding. One participant noted:

“Oh, thought it was fun. It was something different.”

Additionally, *Interpretability and Transparency* emerged as a concern, particularly around automated report generation:

“Maybe...generate a report... what kind of report will it be?”

This comment reflects the user’s desire to understand not just outputs but also the underlying assumptions and processes.

a) System Improvements Following non-Expert Testing:

Based on early-stage feedback, the tool was refined to improve system visibility during message selection and filtering, and to streamline report generation for clearer scope and content prior to testing with humanitarian professionals.

With these improvements in place, the Aegis system was subsequently evaluated via the aforementioned task analysis study with humanitarian professionals, whose domain expertise provided deeper insight into the tool’s potential for operational use in real-world humanitarian contexts.

B. User Study - Humanitarian Expert users

As discussed previously, in addition to the Tool-Centered Goal deductive coding framework, we conducted inductive open coding on humanitarian expert task analysis transcripts. While several themes—such as Contextual Filtering, Data Quantification, Advocacy, Analysis Burden, Analytical and prior AI experience, and AI value—emerged directly from participant responses, many could be traced back to the structure and content of the interview questions themselves. Others arose organically through the analysis process. These overlapping themes were refined into 10 final codes, each with defined criteria summarized in Table 1.

TABLE I
USER-CENTERED GOAL CODES

Code	Definition
<i>Data Integrity Concerns</i>	Users worried about repeated content skewing results or confusing insight counts
<i>Desire for Control & Specificity</i>	Users want more refined filtering to control AI outputs
<i>Desire for Quantitative Support</i>	Users want countable metrics to support interpretations
<i>Emergent Insight Discovery*</i>	AI surfaces surprising or unanticipated patterns
<i>Humanitarian Application*</i>	System used to inform advocacy or programmatic actions
<i>Metadata Limitations</i>	System usability is constrained by metadata gaps
<i>Reduced Human Burden*</i>	System replaces labor-intensive processes
<i>Trust in AI*</i>	Users trust AI but want human or quantitative validation
<i>User Readiness & Training*</i>	Baseline awareness or comfort with AI impacts use
<i>Value-Added AI*</i>	AI enhances traditional tools but is not standard practice

* Indicates that the code was derived from an interview question.

These codes applied were to transcripts using aforementioned prompt engineering, and outputs reviewed for relevance.

a) *Results – Tool-Centered Goal: System Performance and Technical Capability*: Humanitarian experts evaluated the Aegis platform as a promising augmentation to traditional qualitative workflows, particularly due to its technical capabilities related to AI-enhanced classification, geocoding, and

visual analytics. Open thematic analysis revealed participants frequently noted the platform's ability to reduce manual analytic burden. One user explained, *"I don't have to go through hundreds of posts manually; it's all summarized for me in real time."* This suggests Aegis can streamline workflows by automating labor-intensive tasks such as message filtering and topic summarization.

Participants valued the platform's AI-enabled taxonomy classification for surfacing overlooked patterns and guiding analysis. For example, one expert observed, *"I'm also a little bit surprised that health and education were so underrepresented... That gives me something to follow up on."* Others appreciated its potential to support user-driven analysis, while also calling for more control and specificity in queries: *"Just knowing that messages are within that category is not always enough—I'd want to filter further by urgency or sentiment."*

Limitations in system metadata were also identified as obstacles to analysis. As one user stated, *"It looks like most of them probably don't have a location, which makes it hard to map needs."* This concern was echoed in suggestions to strengthen geospatial tagging and metadata extraction to enhance situational awareness. Finally, experts expressed interest in combining qualitative outputs with quantitative data: *"Would I be able to ask it to, like, list the number of people mentioning this?"* This demonstrates a desire to integrate descriptive analytics with inferential insights.

b) *Results – User-Centered Goal: Humanitarian Insight and Decision-Making:* Aligned with the tool's user-centered goal, humanitarian experts from IOM generally found Aegis to be usable, engaging, and applicable to operational needs. As reflected in the deductive coding framework, the interface was described as intuitive and easy to navigate. One participant remarked, *"I think it was pretty user friendly overall."* The system also fostered active emotional engagement, with another user noting, *"That's super interesting... Very interesting."*

However, users noted areas for cognitive improvement, particularly the need for clearer message context: *"...seeing whatever was asked before this or the conversation thread, I think would be helpful..."* In parallel, concerns about transparency and interpretability surfaced, where participants wanted greater traceability between AI outputs and source data: *"...taking that and looking at the messages... I would want a clear link back to where the AI is pulling from."* Another suggested, *"maybe a preset sentence or two on how this was pulled would make me trust it more."*

The tool was perceived as valuable for real-time humanitarian insight generation. Experts indicated that Aegis could help surface emerging needs, especially when navigating large volumes of user-generated content. These findings suggest that Aegis supports a decision-making environment in which regional professionals can engage with dynamic, public discourse to inform data-driven interventions. Its visual interface, summarization, and AI-driven categorization were seen as well-suited for rapid thematic discovery and prioritization—key components in early humanitarian response.

IV. CONCLUSION, LIMITATIONS, AND RECOMMENDATIONS

This study tested the Aegis platform, which combines large language models (LLMs) and visual analytics to support humanitarian decision-making during forced displacement events. Through a two-phase user study involving both non-expert and humanitarian professional participants, we gained valuable insights into the tool's usability, technical capabilities, and applicability in operational contexts.

Key Findings

1. Usability and Learnability: Initial testing revealed that Aegis was largely intuitive and accessible, requiring minimal training. Participants appreciated the clear layout and ease of use, although minor interface issues—such as persistent visual elements—introduced occasional confusion.

2. Technical Capabilities: Humanitarian professionals identified Aegis as a strong augmentation to traditional analytic methods. LLM-based classification and summarization were effective in reducing manual effort and helped surface relevant themes across large social media datasets. However, gaps in metadata—especially in location information—limited the system's capacity for full situational awareness.

3. Trust and Transparency: Users expressed the need for better transparency in how outputs were generated. There was strong interest in tracing summaries back to source content and understanding how LLM-generated reports were derived.

4. Decision Support Potential: Aegis was seen as a promising tool for surfacing emerging needs, tracking community concerns, and aiding prioritization—particularly in fast-moving crises where real-time insight is critical.

The use case study performed allowed IOM employees to navigate message data with visual data analytics quickly. Previous data collection methods included manually scrolling through chat histories to read conversations, which proved to be time-consuming. Aegis allows users to analyze larger amounts of data more rapidly, and with built-in exporting and sharing features, Aegis provides the capabilities to readily communicate data insights and trends. Participants emphasized the importance of combining qualitative narratives with quantitative summaries.

Limitations

This study has several limitations. The system was tested with only four participants, limiting generalizability. While Aegis was found to be largely intuitive, minor interface issues caused occasional confusion. Gaps in metadata—especially location information—hindered full situational awareness, and users expressed a need for greater transparency in how LLM-generated outputs were produced. Additionally, participants highlighted the importance of integrating qualitative insights with quantitative metrics, which remains an area for future development.

Recommendations and Next Steps

Based on these findings and the limitations identified during the study, we offer the following concrete recommendations to guide future work:

- **Enhance AI Capabilities:** Expand geocoding models to support more granular location tagging (e.g., neighborhoods, facilities), and fine-tune LLMs for humanitarian-specific prompt contexts.
- **Support Model Comparisons:** Evaluate updated versions of ChatGPT and compare them with models such as Google Gemini, Meta's LLaMA, and Microsoft Copilot to optimize performance for humanitarian analysis.
- **Improve Transparency and Interpretability:** Provide clear explanations of AI-generated outputs and link reports to original messages to strengthen user trust and credibility.
- **Refine UI Design:** Improve UI elements to better handle multi-resolution displays and mobile deployments. Add features for message quantification, sentiment filtering, and urgency tagging to enhance analytic flexibility.
- **Scale for Larger Data Sets:** Investigate Aegis's performance under high-volume data ingestion scenarios and adapt models and infrastructure to support scalable, real-time processing.
- **Support Mixed-Initiative Exploration:** Extend the tool's interaction model to better support mixed human-AI workflows by integrating user feedback into iterative LLM responses and system recommendations.
- **Expand Evaluation and Real-World Deployment:** Conduct broader evaluations with humanitarian users from multiple regions and organizations. Deploy Aegis during active humanitarian responses to validate effectiveness and identify context-specific needs.

Conclusion

Aegis shows strong promise as a visual analytics platform capable of enhancing humanitarian situational awareness through LLM-powered social media analysis. Feedback from humanitarian professionals underscores its potential to bridge the gap between unstructured public discourse and structured decision-making. With further refinement and expanded testing, Aegis could serve as a valuable tool for rapid insight generation in future displacement crises.

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